



Emma Ricart, PhD

Data Engineer | Bioinformatics | AI

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- 📍 Lausanne, Switzerland

Programming

- ▶ Python
- ▶ R
- ▶ Bash
- ▶ Nextflow
- ▶ Java
- ▶ Perl
- ▶ Snakemake

Server side

- ▶ Docker & Conda
- ▶ MongoDB & PostgreSQL
- ▶ Tomcat
- ▶ Node.js
- ▶ REST API
- ▶ HPC

Frameworks/Libs

- ▶ Pandas & Numpy
- ▶ Matplotlib
- ▶ Scikit learn
- ▶ Django

Profile

Bioinformatics data engineer and scientific software engineer with more than 10 years of experience developing scalable data pipelines, machine learning models, and research software for genomics and mass spectrometry. My work combines data engineering, statistical modeling, and software development to transform complex biological datasets into reproducible computational solutions. I enjoy designing maintainable software and promoting reproducible research through modern engineering practices.

Work experience

Research Scientist

11/2024 - 03/2026

Gulhan Lab

📍 Harvard Medical School | Mass General Hospital - Boston
(Remote)

- Designed scalable and reproducible bioinformatics workflows for preprocessing, integration, and analysis of large-scale genomic sequencing data
- Developed machine learning models and statistical analyses to infer gene expression patterns and classify tumor subtypes using circulating cell-free DNA (cfDNA)
- Applied data engineering and computational methods to support liquid biopsy applications for cancer diagnosis, disease monitoring, and patient stratification
- Mentored junior researchers in workflow development, software engineering best practices, and reproducible computational research

⚙️ *Python – Machine Learning – Snakemake – SLURM – cfDNA genomics – Git*

Bioinformatician

02/2021 - 02/2024

Bassani Lab (Prof. Michal Bassani)

📍 AGORA Cancer Research Center - Lausanne

- Developed a workflow for large-scale identification and quantification of HLA/HLAII peptides from mass spectrometry experiments
- Implemented an ETL process for the creation of an immunopeptidomics data warehouse which included over 1 million peptides identified with the workflow
- Used the collected data to create a scoring system for the prioritization of neoantigens, directly impacting the clinical trials

⚙️ *Python – R – Bash – Nextflow – HPC – Immunopeptidomics – Mass spectrometry – Data Visualization – Biological Data Analysis – Docker – Software development – Git*

Ph.D. Student

09/2015 - 09/2019

Proteomic Informatics Group (Frederique Lisacek)

📍 Swiss Institute of Bioinformatics (SIB) - Geneva

- Developed rBAN, a java package and tool for the retro biosynthesis of non-ribosomal peptides
- Developed NRPro, a tool for the identification and annotation of tandem mass spectra from peptidic natural products
- Worked with SMILES format and graph structures for the substructures search and the fragmentation algorithms used by the tools
- Build a NoSQL database for the backend of NRPro integrating data from Norine, ChEBI and Natural Products Atlas

⚙️ *Java – R – Maven – MongoDB – AngularJS – Mass spectrometry – Peptidomics – Data Visualization – Biological Data Analysis – Software development – Git*

Junior Bioinformatician

02/2015 - 09/2015

The Chuang Lab (Prof. Jeffrey H. Chuang)

📍 JAX Cancer Center - Connecticut

- Analyzed iCLIP data from ENCODE project to study RNA-protein binding sites

Development tools

- ▶ Maven
- ▶ Git
- ▶ Jenkins

Hard-skills

Data analysis 10+ yrs

Software development 5+ yrs.

Soft-skills

- ▶ Learning agility
- ▶ Teamwork
- ▶ Communication skills
- ▶ Critical thinking
- ▶ Enthusiasm
- ▶ Initiative
- ▶ Trustworthiness
- ▶ Presentation skills

Languages

English C1

French B1

Spanish & Catalan C2

⚙️ *Python – R – Perl – Bash – HPC – Biological Data Analysis – Statistics – Data Visualization – Transcriptomics*

Bioinformatics researcher

09/2014 - 02/2015

Computational Genomics group (Prof. Eduardo Eyras)

📍 PRBB - Barcelona

- Analyzed and quantified iTRAQ mass spectrometry data from the TCGA project to validate findings identified at RNA level

Master's Student

02/2014 - 06/2014

Computational Proteomics group (Fredrik Levander)

📍 Lund University - Sweden

- Identified and quantified of protein variants using a combination of shotgun proteomics and DIA LC-MS

Education

Ph.D. in Bioinformatics

09/2015 - 09/2019

Swiss Institute of Bioinformatics (SIB)

University of Geneva

Master's degree in Biomedical engineering

09/2012 - 09/2014

University of Barcelona

Polytechnic University of Catalonia

Lund University

Bachelor's degree in Biotechnology

09/2007 - 09/2011

University of Vic - Central University of Catalonia

Awards

Winner Best Lighting Talk Award

2018

SIB Days Awards 2018

SIB PhD Fellowship Laureate

2015

SIB Fellowship Program 2015

Relevant publications

Ricart, Emma, Maude Pupin, Markus Muller, and Frédérique Lisacek. Automatic annotation and dereplication of tandem mass spectra of peptidic natural products. *Analytical Chemistry*, 92(24):15862–15871, 2020.

Ricart, Emma, Valérie Leclère, Areski Flissi, Markus Mueller, Maude Pupin, and Frédérique Lisacek. rban: retro-biosynthetic analysis of nonribosomal peptides. *Journal of cheminformatics*, 11(1):1–14, 2019.

Markus Müller, Florian Huber, Marion Arnaud, Anne Kraemer, Altimiras, Emma Ricart, Justine Michaux, Marie Taillandier-Coindard, Johanna Chiffelle, Baptiste Murgues, Talita Gehret, et al. Machine learning methods and harmonized datasets improve immunogenic neoantigen prediction. *Immunity*, 56(11):2650–2663, 2023.